17-7 PH Stainless Steel, TH1050, bar and forgings

Categories: Metal; Ferrous Metal; Stainless Steel; Precipitation Hardening Stainless; T S10000 Series Stainless Steel

Material Notes: Properties are measured in the longitudinal direction.

Processing: TH1050 - heated to austenitic range, 1040°C (1900°F), and water quenched. Reheated to 760°C (1400°F) to precipitate carbides. Carbides cause the austenite transform to martensite after cooling below 15°C (60°F). Tempered at 565°C (1050°)

Applications: high strength high temperature applications, chemical processing equipment, heat exchangers, power boilers, superheater tubes

Corrosion Resistance: 17-7 PH is suitable for use in fresh water, industrial and marine atmospheres, and mild chemical and oxidizing environments. 17-7 PH should not be used in salt water or reducing environments.

Key Words: UNS S17700, double treatment alloy, TH1050, bar and forgings, 17-7PH, 17-7 PH, 17/7PH, 17/7 PH, Precipitation Hardening

Vendors: **Click here** to view all available suppliers for this material.

Please click here if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	Metric	English	Comments
Density	7.80 g/cc	0.282 lb/in³	

Mechanical Properties	Metric	English	Comments
Hardness, Rockwell C	38	38	
Tensile Strength, Ultimate	1170 MPa	170000 psi	
Tensile Strength, Yield	965 MPa	140000 psi	
Elongation at Break	6.0 %	6.0 %	
Reduction of Area	25 %	25 %	
Modulus of Elasticity	204 GPa	29600 ksi	

Electrical Properties	Metric	English	Comments
Electrical Resistivity	0.0000830 ohm-cm	0.0000830 ohm-cm	

Thermal Properties	Metric	English	Comments
CTE, linear	11.0 µm/m-°C @Temperature 0.000 - 100 °C	6.11 µin/in-°F @Temperature 32.0 - 212 °F	
	11.6 µm/m-°C @Temperature 0.000 - 315 °C	6.44 µin/in-°F @Temperature 32.0 - 599 °F	
Specific Heat Capacity	0.460 J/g-°C @Temperature 0.000 - 100 °C	0.110 BTU/lb-°F @Temperature 32.0 - 212 °F	
Thermal Conductivity	16.4 W/m-K @Temperature 100 °C	114 BTU-in/hr-ft²-°F @Temperature 212 °F	
	21.8 W/m-K @Temperature 500 °C	151 BTU-in/hr-ft²-°F @Temperature 932 °F	
Melting Point	1400 - 1450 °C	2550 - 2640 °F	
Solidus	1400 °C	2550 °F	
Liquidus	1450 °C	2640 °F	

Component Elements Properties	Metric	English	Comments
Aluminum, Al	0.75 - 1.5 %	0.75 - 1.5 %	
Carbon, C	<= 0.090 %	<= 0.090 %	
Chromium, Cr	16 - 18 %	16 - 18 %	
Iron, Fe	70.6 - 76.8 %	70.6 - 76.8 %	as balance
Manganese, Mn	<= 1.0 %	<= 1.0 %	
Nickel, Ni	6.5 - 7.75 %	6.5 - 7.75 %	
Phosphorus, P	<= 0.040 %	<= 0.040 %	
Silicon, Si	<= 1.0 %	<= 1.0 %	
Sulfur, S	<= 0.040 %	<= 0.040 %	